MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 8932 1/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: UHF, beam power tetrode, pulse amplifier, transmitting
See figure 1
Mounting position: Tube axis vertical, either end up
Weight: 38 pounds nominal

ABSOLUTE RATINGS:
(Anode pulsed amplifier, class B, see note 4)

Parameter: Unit:	Ef Vac (see note 1)	ebb kv (see note 2)	Ebb kVdc	ec2 kv (see note 3)	Ec1 Vdc	ib a	Ib mAdc	ic1 a (see note 5)	Ic1 mAdc	ic2 a
On time = 15 μs										
Maximum:	1.05(dc) 1.05(60 Hz) 1.05(400 Hz)	45	---	2.2	-400	80	320	15	60	15
Minimum:	---	---	---	---	---	---	---	---	---	---
On time = 60 μs										
Maximum:	1.05(dc) 1.05(60 Hz) 1.05(400 Hz)	30	20	2.2	-400	60	1,200	5	90	5
Minimum:	---	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	---	---	---	---	---	---	---	---	---	---

ABSOLUTE RATINGS:
(Anode pulsed amplifier, class B, see note 4)

Parameter: Unit:	Ic2 mAdc	Pi kW	Pp kW	tk sec (see note 6)	Cooling --- (see note 7)	Environmental requirements --- (see note 9)	Fl MHz	Fault protection (see note 10)	Duty ---	If(surge) A (see note 6)
On time = 15 μs										
Maximum:	60	16	14	---	---	---	550	---	0.004	550
Minimum:	---	---	---	90	---	---	---	---	---	---
On time = 60 μs										
Maximum:	90	16	14	---	---	---	550	---	0.02	550
Minimum:	---	---	---	90	---	---	---	---	---	---
TEST CONDITIONS:	---	---	---	---	Note 8	---	---	---	---	---

GENERAL:

Qualification - Required
Holding period - t = 96 hours

1/ Formerly identified as tube type 4666 (see note 29)

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
1155	Anode water pressure differential	14	No voltages applied	0.65	II	PPD	---	10.0	psi
1155	Grid (1) water pressure differential	14	No voltages applied	0.65	II	GPD	---	6.0	psi
1155	Grid (2) water pressure differential	14	No voltages applied	0.65	II	SPD	---	8.0	psi
1155	Filament water pressure differential	14	No voltages applied	0.65	II	FPD	---	8.0	psi
1155	Cathode water pressure differential	14	No voltages applied	0.65	II	CPD	---	8.0	psi
1261	Electrode voltage (grid)	-	Ebb = 15 kVdc; ec2 = 2 kv; Ec1/Ib = 2.7 a; Ef = 0.95 V, 60 Hz; prr = 300 ± 5; tp = 60^{+5}_{-0} µs	0.65	II	ec1	---	-320	v
1301	Filament current	-	Ef = 0.95 V, 60 Hz	0.65	II	If	460	530	A
---	Peak power output	1, 17, 19, 27, 28, 31	Ef = 0.95 V, 60 Hz; Ebb = 15 kVdc; ec2 = 1.8 kv; Ec1 = -250 ± 10 Vdc; pd/Ib = 60 a; prr = 300 ± 5; tp = 60^{+5}_{-0} µs; F = 400 MHz (min)	0.65	II	po Ic1 Ic2 Power gain	225 10 10 50	290 --- --- ---	kw mAdc mAdc ---
---	Emission check	30	Peak power output; Ef/If = 450A, 60 Hz; ec2 = 2 kv (max); ib = 60 a (max)	0.65	II	po	225	---	kw
---	Perveance	1, 18	Ef = 0.95 V, 60 Hz; ec1 = ec2 = 200 v; tp = 1.0 µs (min); prr = 10 (min)	0.65	II	is	54	---	a
1256	Electrode current (grid)	20, 26	Ef = 0.95 V, 60 Hz; Eb = 6.5 kVdc; Ec2 = 600 Vdc; Ec1/Ib = 1.0 Adc; t = 5 minutes	0.65	II	Ic1	---	-50	µAdc
1256	Electrode current (screen)	-	Ef = 0.95 V, 60 Hz; Ec2 = 600 Vdc; Ec1/Ib = 1.0 Adc; Eb = 6.5 kVdc	0.65	II	Ic2	-25	25	mAdc
---	RF aging time	15	Peak power output except, Ebb = 16.5 kVdc; Ib = 1.2 Adc; ec2 = 2.3 kv; Ec1 = -310 ± 40 Vdc (notes 19, 27, 28, and 31 do not apply)	0.65	II	t	10	---	min

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1 -Continued								
---	Input resonant frequency	21		0.65	II	Fi	230	245	MHz
---	Output resonant frequency	21		0.65	II	Fo	240	245	MHz
	Quality conformance inspection, part 2								
1331	Direct-interelectrode capacitance	22	Use special shield adapter DESC Drawing No. 66060, or equivalent	---	---	Cg2k Cg1p	10,000 ---	18,000 0.15	pF pF
	Quality conformance inspection, part 3								
---	Life test	-	Group S	---	---	---	---	---	---
---	Initial life	1, 11, 13, 16, 17, 32	Ef = 0.95 V, 60 Hz; ebb = 30 kv; ec2 = 2 kv (max); Ec1 (note 16); pd/Ib = 165 mAdc; po = 750 kw (min); F = 225 ±15 MHz; t_p = 10 ±1 µs; prr = 300 ±5; specified life = 1,000 hours	---	---	---	---	---	---
---	Initial life-test end point:								
---	Peak power output	-	Initial life	---	---	po	750	---	kw
---	Multiple short life (1)	12, 33	Initial life except, po = 825 kw (min); t = 50 hours	---	---	---	---	---	---
---	Multiple short life-test end point (1):								
---	Peak power output	-		---	---	po	825	---	kw
	Periodic-check tests								
---	Vibration	23, 24		---	---	---	---	---	---
---	Shock	23, 25		---	---	---	---	---	---
---	Vibration and shock test end point:								
---	Peak power output	-		---	---	po	225	---	kw

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NOTES:

1. Because the filament voltage, when operated near the maximum value, provides emission in excess of any requirements within tube rating, during life, the filament voltage shall be reduced to a value that will give adequate, but not excessive emission. Careful attention to maintaining the value consistent with adequate emission will result in conserving the life of the tube. The filament voltage shall be measured at the respective liquid coolant connections on the tube side of the threads. This procedure is essential in order to accurately measure the filament voltage. At 400 Hz some heating of the filament leads and rf cathode terminal (cathode header) occurs; this conditions is not detrimental to tube operation or tube life.

2. Peak positive anode supply voltage. The peak pulse value is defined as the maximum value of a straight line through the average of the fluctuations over the top portion of the pulse. The magnitude of any spike, on the anode voltage pulse, shall not exceed the peak pulse value by more than 4,000 volts, and the duration of any spike (measured at a point 70 percent above the peak pulse value) shall not exceed 2 μs. In the absence of rf output voltage, the peak pulse supply voltage may be exceeded by 20 percent but not more than 10 consecutive pulses. A pressurized output cavity is required, for operation at the maximum voltage specified herein, to prevent external flashover at the insulating bushing. (See note 9.)

3. Peak positive screen voltage. The peak pulse value is defined as the maximum value of a straight line through the average of the fluctuations over the top portions of the pulse. The magnitude of any spike, on the grid(2) voltage pulse, shall not exceed the peak pulse value by more than 250 volts and the duration of any spike, (measured at a point 70 percent above the peak pulse value) shall not exceed 2 μs.

4. On time is defined as the sum of the durations of all the individual pulses which occur during any 3,000 μs interval.

5. Peak rectified grid(1) current.

6. Minimum time to reach operating filament voltage shall be 30 seconds. After operating filament voltage has been obtained, a minimum of 90 seconds heating time is required before other electrode voltages can be applied. Filament current shall not exceed 550 amperes, even momentarily.

7. Cooling requirements.

   (a) Air-cooling. Forced-air-cooling of the ceramic bushing at the grid(1) seal and at the anode seal may be required in order to limit the temperature of the ceramic bushing at either seal to the specified maximum value of 150° C. Under such conditions, provisions shall be made for blowing air at the ceramic bushing through suitable openings in the coaxial-cylinder cavity circuit.

   (b) Liquid cooling. Liquid-cooling of the filament block, dc cathode block, grid(1) block, grid(2) block, and anode, is required. When the environmental temperature is above 0° C, the coolant may be water. The liquid flow shall start before application of any voltages and perferably shall continue for several seconds after removal of all voltages. Interlocking of the liquid flow through each of the cooled elements with all power supplies is recommended to prevent tube damage in case of failure of adequate liquid flow. Liquid flow shall be as shown in table I.

TABLE I. Liquid flow.

Coolant course	Waterflow		Maximum pressure differential 1/ for typical flow (psi)
	Absolute minimum (gpm)	Typical (gpm)	
With water:			
Through filament block	0.5	0.8	8
Through dc cathode block	0.5	0.8	8
Through grid(1) block	0.5	0.8	6
Through grid(2) block	0.5	0.8	8
Through anode in direction shown on figure 1:			
For anode dissipation up to 11 kw (average)	8	10	10
For anode dissipation of 14 kw (average)	10	12	15

1/ Directly across cooled element.

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NOTES: -Continued

8. The waterflow to the anode cooling connection shall be 8 $^{+1}_{-0}$ gpm. The minimum waterflow to the filament, filament-cathode, grid(1) cooling connections, shall be 0.5 gpm. Waterflow shall start before the application of any voltages, and shall be continued for at least 6 seconds after removal of all voltages. The outlet water temperature shall not exceed 70° C.

   Emergency conditions. Simultaneous interruption of cooling pump and tube electrode power is permissible.

9. Environmental requirements.

External gas pressure (absolute) 1/	60 2/ psi (max)
Gage pressure at any inlet	100 psi (max)
Metal-surface temperature	100° C (max)
Ceramic-bushing temperature	150° C (max)
Outlet liquid temperature (any outlet) for water	70° C (max)
Storage temperature 3/	-65° C (min)

   1/ Absolute pressure is the sum of the atmospheric pressure and gage pressure.
   2/ This pressure is related to the output cavity pressurization as required to prevent corona or external flashover.
   3/ The tube coolant ducts shall be free of liquid before storage or shipment of the tube to prevent damage from freezing.

10. High-speed electronic protective devices shall be used to prevent energy from the anode-power supply, grid(1) power supply, and from the grid(2) power supply from permanently damaging the tube and its circuitry in the event of abnormal operation. A suitable high-speed electronic protective device shall be capable of deactivating the various pulse sources supplying the tube during the inter-pulse interval following the occurrence of a fault in any single pulse, and shall be capable of meeting the requirements of the aluminum foil simulated fault tests. Thus, all pulses subsequent to a "fault" pulse shall not be applied until the device is reset. (A fault is a condition in which the peak pulse currents rise 20 percent or more above the normal peak current level.) Additional protection against damage from an anode-to-grid(2) fault shall be provided to prevent the grid(2) voltage from rising to a value in excess of 3,000 volts during the fault pulse. Minimum reset time shall be 1 second.

11. This requirement shall be waived for manufacturers qualified to tube type 8932 (see note 29).

12. The following sampling plan shall be used:

Lot size	Sample	Sample size	Cumulative sample size	See 10.1.2 of MIL-STD-105 Accept	Reject
1	First	1	1	0	1
	Second	0	0	-	-
2	First	1	1	0	-
	Second	1	2	1	2
3 to 10	First	1	1	0	-
	Second	2	3	1	2
Over 10	First	3	3	0	2
	Second	3	6	1	2
	Second (see note)	3	6	2	3

NOTE: The second sample criteria applies only if the first sample from the previous lot had zero failures.

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NOTES: -Continued

13. Water system requirements. In order to minimize maintenance, it is recommended that the water cooling system be of the closed type, utilizing water of acceptable quality as specified herein, to prevent excessive deposition, corrosion, and electrolysis. Deposition of material not only restricts waterflow, but also prevents proper transfer of heat from the tube elements to the cooling water. Corrosion and electrolysis may destroy the tube elements, ducts, and fittings. Cooling water of acceptable quality shall have a specific resistivity in excess of 1 megohm-cm.

14. The waterflow to the anode cooling connection shall be 10 gpm. The waterflow to the filament, filament-cathode, grid(1) and grid(2) cooling connections, shall be 0.8 gpm. Pressure drops are measured at the tube terminals.

15. The tube shall operate continuously under the specified test conditions.

16. The grid(1) voltage (Ec1) is adjusted for class B operation. Class B operation is defined as a negative Ec1 voltage which is between one-fifth and one-eighth of the ec2 voltage.

17. Circuit adjustments are permissible. The rectified rf output pulse length (tp) and the pulse repetition rate (prr) shall be as specified herein. Peak output power is calculated by dividing the average output power by the duty factor. Duty factor is defined as the product of pulse duration and the repetition rate. The average output power is measured by a calorimetric method using a dummy water load, RCA Model MI 19198-A, or equivalent. The specified test duration (t) shall be for continuous operation. Minimum output circuit bandwidth as checked by cold probe techniques shall be 12 MHz to the 3 dB points and 20 MHz to the 6 dB points.

18. With the anode connected to the cathode and grid(2) connected to grid(1) apply a partial sinusoidal voltage of 200 v peak to the grid(1). Emission current shall be read on an oscilloscope connected across a noninductive resistor in series with the grid(1)-grid(2) cathode circuit or by means of a current transformer.

19. The rf input power is measured using a calibrated directional coupler. This power is the difference between the incident and reflected power as sampled by the directional coupler. Coupler calibration is accomplished by the calorimetric method, rf substitution method or other approved methods. Power gain is calculated by dividing the rf output power by the rf input power. The grid circuit shall be adjusted for lowest VSWR in the drive line. Specification readings can not be taken unless reflected power is less than 10 percent of the incident power.

20. With conditions as specified herein, grid(1) current (I_1) shall be recorded after specified time. All voltages shall be removed except filament voltage and grid(1) voltage, and then grid(1) leakage current (I_2) shall be recorded. Calculate Ic from the formula: $Ic = I_1 - I_2$.

21. The following procedure (or equivalent) shall be used for measuring input and output resonant frequency.

    (a) Set up equipment as in figure 2. Be sure tube is making good contact with cavity.

    (b) Adjust the frequency of rf generator to obtain a resonant peak on the standing-wave indicator.

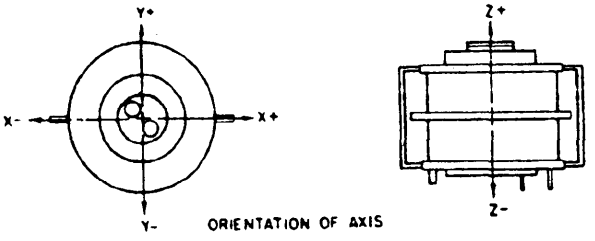
    (c) The resonant frequency as read on frequency meter shall be within limits specified herein. (The input cavity and output cavity shall be as shown on figures 3 and 4.)

22. Each element not under test shall be tied to ground. Two tubes per month shall be tested. Failure of one or more tubes will require 100-percent testing. Two tubes per month may be resumed after testing of 20 consecutive tubes without failure.

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NOTES: -Continued

23. Vibration and shock tests shall be performed using vibration-shock fixture (DESC Drawing No. 67031), or equivalent. Tests shall be performed with no voltages applied.



ORIENTATION OF AXIS

Post vibration and shock need only be performed once (at the conclusion of both vibration and shock tests).

24. Vibration.

(a) The tube shall be vibrated in the X, Y, and Z axes at the following conditions:

Frequency (Hz)	Double amplitude (inches)
5 to 55	.060

(b) Frequency shall be swept logarithmically from 5 to 55 to 5 Hz in 5 minutes minimum. Two sweeps shall be made.

(c) The tube shall be vibrated in the X, Y, and Z axes at the following conditions:

Frequency (Hz)	Double amplitude (inches)	Duration (hours/axis)
50	.060	12

(d) Tubes which do not comply with the post-vibration limits shall be considered failures.

25. Shock. Test in accordance with MIL-STD-202, method 213, test condition A, except as follows:

(a) The peak acceleration of the shock shall be 15 G ± 10 percent.

(b) The velocity change (V_i) = 3.39 ft/sec.

(c) Tubes which do not comply with post-shock limits shall be considered failures.

(d) This is not a destructive test.

26. This test is to be the first test performed at the conclusion of the holding period.

27. The anode current pulse shall be examined for evidence of pulse droop. No pulse droop attributable to tube emission shall be permitted.

28. The tube shall operate under power output conditions for 1 hour without a trip-out. Measured data to be taken at the end of this 1-hour time period. A trip-out is defined as an increase in plate current in excess of 20 percent above normal level.

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NOTES: -Continued

29. Tube type 8932 shall be used to replace tube types 4666 and 6952 in AN/SPS-40A, -40B, -40C, and -40D applications. Tube type 6952 shall be used in AN/SPS-29, -32, and -40 applications. Users should evaluate tube type 8932 before using it as a replacement in other applications.

30. Emission check. Test conditions and limits for this test shall be as specified. However, the following alternate method may be used:

Test conditions	Test limits
Ef/If = 450 A, 60 Hz; ebb = 30 kv, ec2 = 2 kv (max); ec1 = (note 16); RF drive/Ib = 165 mAdc (max) (note 17); Ic1 = 0 to +28 mAdc; F = 225 ±15 MHz; tp = 10 ±1 μs; prr = 300 ±5	po = 825 kw minimum

31. The tube shall have passed the Ic1 limit at the end of 1 hour if Ic1 $\geq$ 15 mAdc. If, at the end of 1 hour, the Ic1 reading is between 10 to 14 mAdc inclusive, then continue the test for an additional hour after which Ic1 shall equal or exceed 10 mAdc.

32. The following life test may be used as an alternate to initial life at the option of the manufacturer:

TEST	NOTES	CONDITIONS	SYMBOL	MINIMUM	MAXIMUM	UNIT
Initial life (alternate)	1, 11, 13, 16, 17	Ef = 0.95 V, 60 Hz; Ebb = 22 kVdc (max); ec2 = 2 kv (max); Ec1 = note 16; pd/Ib = 0.50 Adc; po = 200 kw (min); F = 200 MHz (min); tp = 100 μs (min); prr = 240 ±5; t = 1,000 hrs	---	---	---	---
Initial life (alternate) test end point Peak power output	-	Initial life (alternate)	po	200	---	kw

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NOTES: -Continued

33. The following life test may be used as an alternate to multiple short life at the option of the manufacturer:

TEST	NOTES	CONDITIONS	SYMBOL	MINIMUM	MAXIMUM	UNIT
Multiple short life (alternate)	12	Same as initial life (alternate) except, po = 225 kw (min); t = 50 hrs (note 11 does not apply)	---	---	---	---
Multiple short life (alternate) test end point: Peak power output	-	Multiple short life (alternate)	po	225	---	kw

User activities:
Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2719)

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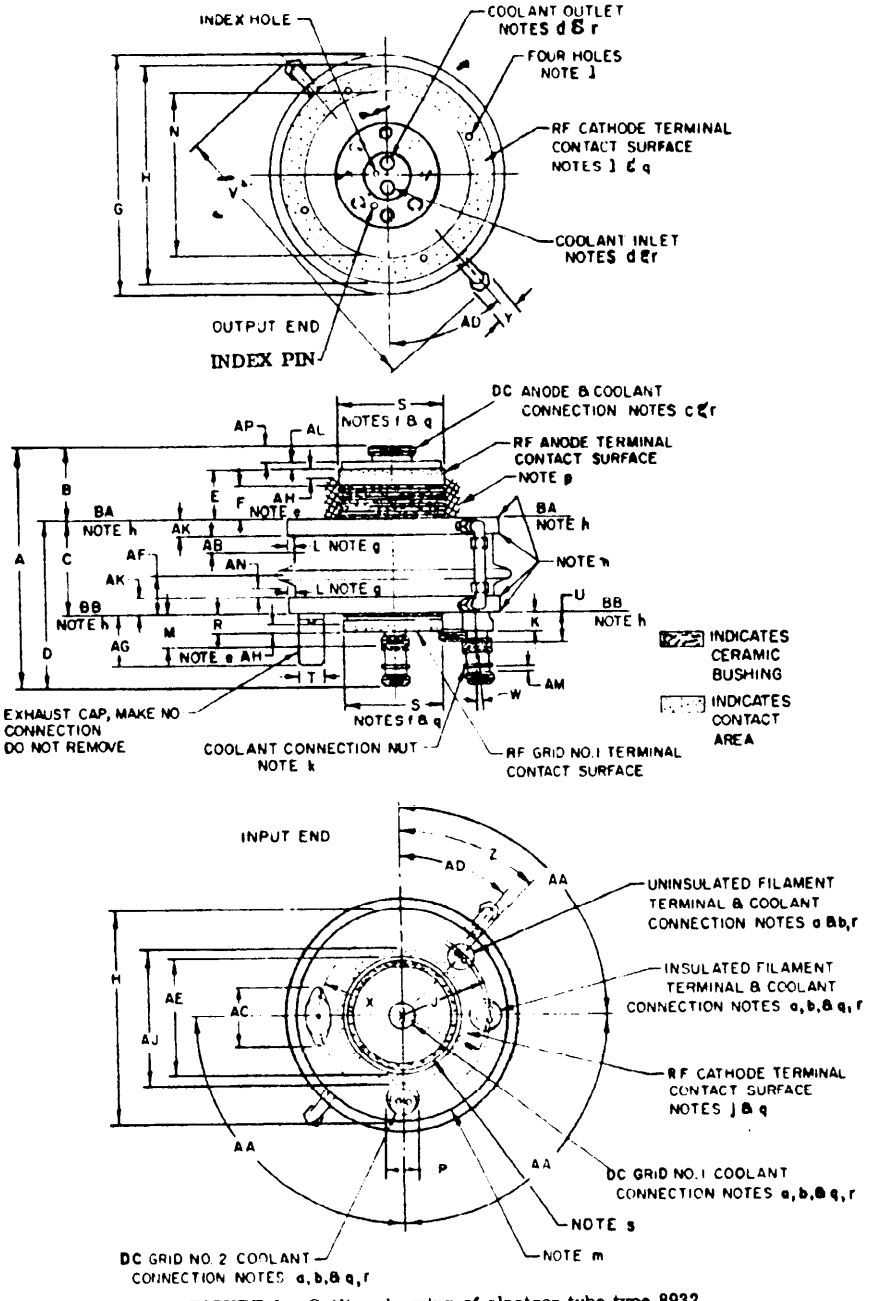


FIGURE 1. Outline drawing of electron tube type 8932.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 1			r
	G-3 Gage check		q
E	1.83 (46.48)	1.93 (49.02)	
R	.59 (14.99)	.69 (17.53)	
S	3.74 (95.00)	3.76 (95.50)	f, q, t
Quality conformance inspection, (periodic check) part 3			
A	8.31 (211.07)	8.93 (226.82)	
C	3.27 (83.06)	3.45 (87.63)	
D	5.68 (144.27)	6.18 (156.97)	
F	1.17 (29.72)		
G		8.75 (222.25)	
H	7.84 (199.13)	7.88 (200.15)	t
K		.59 (14.99)	
L	.13 (3.30)		g, t
M	1.04 (26.42)	1.14 (28.96)	q
P	1.22 (30.99)	1.28 (32.51)	
T	.97 (24.64)	1.12 (28.45)	
U	.83 (21.08)	.95 (24.13)	
V		11.25 (285.75)	
Y		.68 (17.27)	
AB	.50 (12.70)		g
AC	2.09 (53.09)	2.15 (54.61)	
AE		4.06 (103.12)	j
AF	1.38 (35.05)	1.50 (38.11)	
AG		2.00 (50.80)	
AH	.38 (9.65)		e
AJ	4.95 (125.73)		j
AK	.56 (14.22)	.63 (16.00)	
AL		.28 (7.11)	
AM	.09 (2.29)		
AN	.25 (6.35)		g
AP	.52 (13.21)	.60 (15.24)	
	Reference dimensions		
B	2.69 (68.33)		q
N	6.00 (152.40)		l
J	RAD 3.25 (82.55)		q
W	3°		q
X	6.25 (158.75)		q
Z	45°		q
AA	90°		q
AD	41°		q

FIGURE 1. Outline drawing of electron tube type 8932 - Continued.

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NOTES:

a. Terminal has 1 inch (25.4)-16 unified thread, class 2A fit, .380-inch (9.65) long, and two holes .258 - .270-inch (6.55-6.86) diameter, spaced .438 inch (11.13) on centers.
b. The holes in the filament, grid No. 1, and grid No. 2 coolant terminal connections, will accept the pins of the plug-and-cylinder combination gage G1, shown on figure 5.
c. Terminal has 1-3/4 inch (44.45)-16 unified extra fine thread, class 2A fit, .380-inch (9.65) long, two holes .508-.522-inch (12.90-13.26) diameter, spaced .688 inch (17.48) on centers, and an index hole .160-inch (4.06) maximum diameter, spaced .344 inch (8.74) from the center of the terminal.
d. The holes in the anode coolant connection shall accept the pins of the plug-and-cylinder combination gage G2, shown on figure 6.
e. Pressure from circuit contacts shall be exerted only over maximum length, AH, of designated contact areas of the anode or grid No. 1 terminals.
f. This diameter dimension is held only over a length of AH.
g. Dimension L applies over length AB and AN, as indicated.
h. The contact surfaces, BA-BA and BB-BB of the rf cathode terminals, shall be parallel within .06 inch (1.52).
j. Contact of the input end, rf cathode terminal, shall not be made at a diameter smaller than 4.06 inch (103.12), nor greater than 4.95 inch (125.73).
k. To prevent excessive stress on the ceramic seal, a 15/16-inch (23.81) open-end wrench shall be used to permit gripping the terminal when removing or tightening the coolant connectors.
l. Contact of the output end, rf cathode terminal, shall not be made at a diameter smaller than 6 inches (152.40). The pressure exerted for this rf contact, shall be limited to that necessary for good electrical contact. The mechanical force for the cavity support and pressure seal shall be made at a diameter not less than 6 inches (152.40). On the output end, rf cathode terminal, there are four equally spaced .188-inch (4.78) diameter holes on a circle having a diameter of 6.75 inches (171.45). These holes are for tube manufacturing purposes only. Attention is called to the existence of these holes so that equipment designers can avoid making a pressure seal or electrical contact at points which are coincident with these holes. Mechanical clamping devices for the output cavity shall be designed so as to exert their clamping force across outer edge of output-header flange
m. Serial number is located on this surface between grid No. 2 and the insulated filament terminal.
n. Corners may be rounded or chamfered, but not to exceed .05 inch (1.27).
p. Keep this 5-inch diameter annular-volume region clear. Do not allow circuit components (conductors or insulators) to protrude into this region.
q. The output end shall accept the output end of gage G3, shown on figure 7. Proper acceptance is obtained when gage is seated on the output end, rf cathode terminal, contact surface. The gage is properly seated when a .015-inch (.38) thickness gage, 1/2-inch (12.70) wide will not enter between the gage and the rf cathode contact surface. The pins on the 1.500-inch (38.10) movable section of the output end of gage G3, shall enter the anode coolant holes such that the height slot represented by the 2.098-inch (53.29) dimensions is below the shoulder of the gage, and the 2.219-inch (56.36) dimension is visible above the shoulder. The input end shall accept the input end of gage G3, shown on figure 7. Proper acceptance is obtained when the gage is seated on the input end, rf cathode terminal surface, and the projecting guide pins of the output end engage the mating holes of the input end. The gage is properly seated when a .015-inch (.38) thickness gage, 1/2-inch (12.70) wide will not enter between the gage and the rf cathode terminal contact surface. The pins on the .875-inch (22.23) movable section of the input end shall enter the grid coolant holes such that the height slot represented by the 2.379-inch (60.43) dimension is below the shoulder of the gage, and the 2.559-inch (65.00) dimension is visible above the shoulder, when gage G3 is properly seated.
   (1) Dimensions B, J, M, W, X, Z, AA, and AD shall vary only within the limits of the gage.
   (2) The centerline through the grid No. 1 coolant connection holes will have a maximum angle of 2 degrees with respect to the centerline of the anode coolant connection holes.
   (3) The axis of the rf anode terminal contact surface is coincident with the axis of the grid No. 1 terminal contact surface, within .05 inch (1.27).
   (4) The axis of the anode and coolant connection is coincident with the axis of the rf anode terminal contact surface, within .025 inch (.64).
r. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.
s. The outside diameter of this ceramic bushing shall not enter the annular region defined by a circle of 3.88 inch diameter (98.55mm) concentric with diameter S of the rf grid No. 1 terminal.
t. Measurement of this dimension shall be made on both input and output ends.
u. The centers of the water course holes, in each connection around the periphery of the tube, shall be on a line through the center of these holes and perpendicular to a line through the center of the tube within ±2 degrees.

FIGURE 1. Outline drawing of electron tube type 8932 - Continued.

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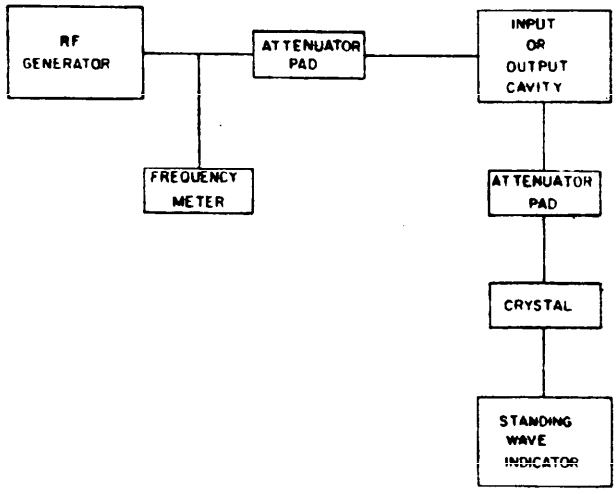
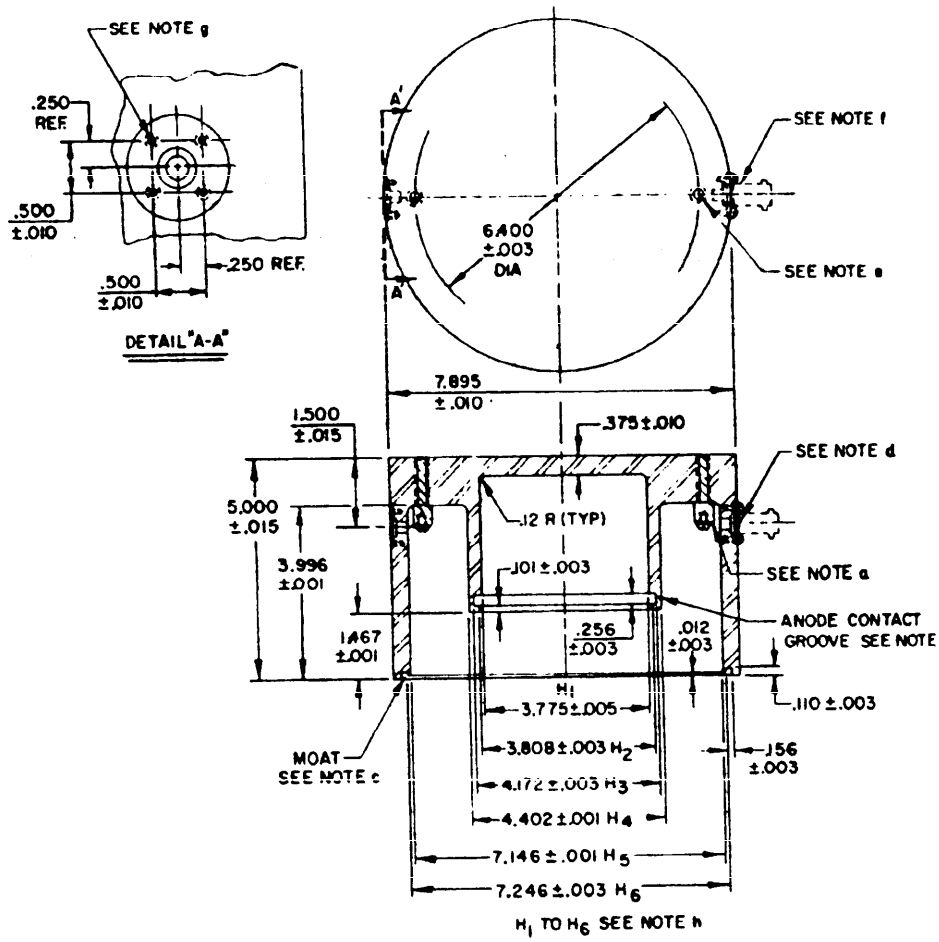


FIGURE 2. Typical block diagram of test circuit.

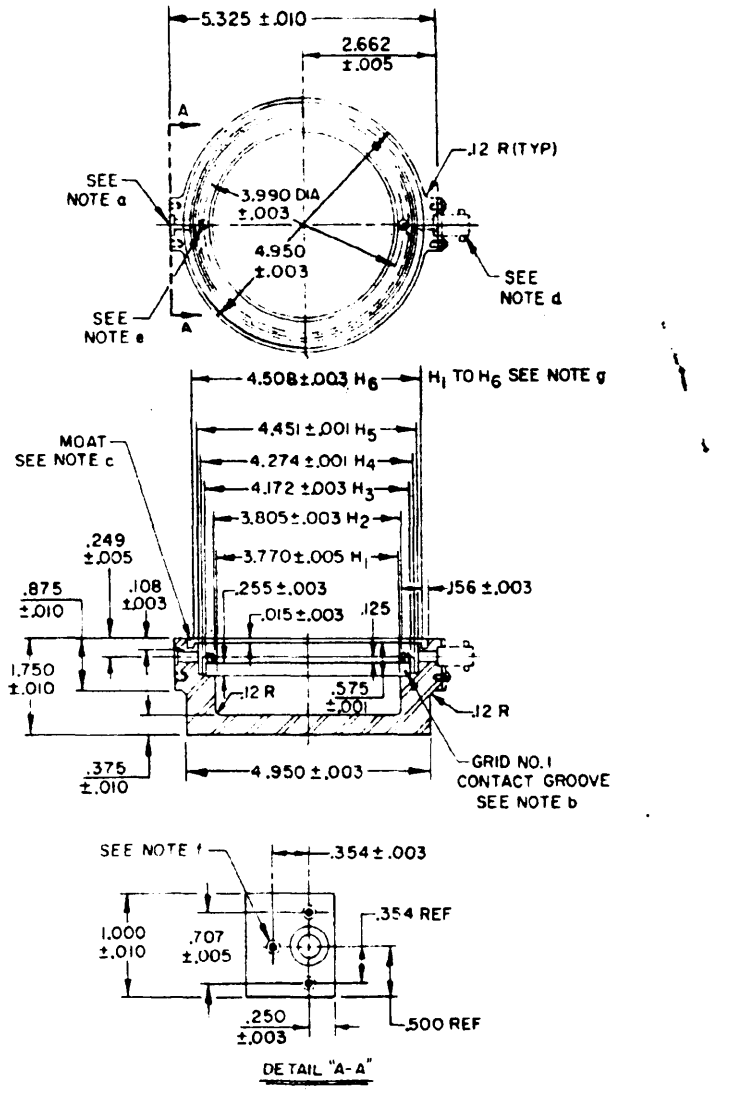
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NOTES:
a. Two studs. 1-1/2-inch long threaded 1/4-inch-20 NC by 1 inch with 3/8-inch-diameter head. Head of stud shall be drilled and tapped 1/4-inch deep for No. 6-32 RH machine screw. Use standard 5/32-inch-flat washer under screw.
b. Anode contact groove shall be filled with the following spring: .020-inch-diameter wire wound to give a .250-inch O.D. at 35 TPI, 14-1/4 inches in length (approx.).
c. Moat shall be filled with a braid to insure continuous braid contact and yet permit firm seating of the cavity on the rf cathode terminal contact surface.
d. Two holes. Drill 7/32-inch diameter (thru), C' bore 3/8- diameter by 1/8-inch deep. Spot face 1-inch diameter by 1/16-inch deep.
e. Two holes. Drill and tap for 1/4-inch-20 NC screws.
f. Location of A-N connector-UG-290/U (two connectors 180 degrees apart).
g. Four holes. Drill and tap for No. 3-48 screws by 1/4-inch deep.
h. Cylindrical holes H_1 to H_6 have axes coincident within .001 inch.
i. Dimensions are in inches.

FIGURE 3. Output cavity required for resonant frequency test.

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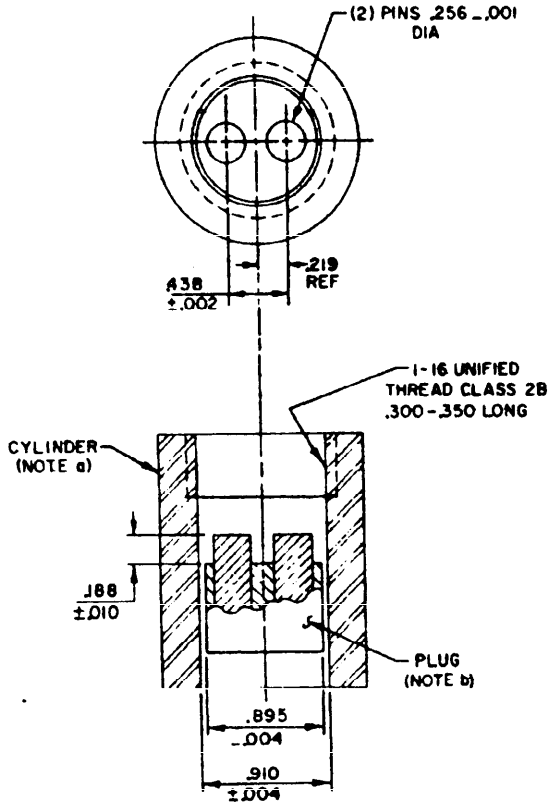
NOTES.

a. Two holes, drill 7/32-inch diameter (thru), C' bore 3/8-inch diameter by 1/16-inch deep.
b. Grid No. 1 contact groove to be filled with the following spring. .020-inch diameter wire wound to give a .250-inch O.D. at 35 TPI, 14-1/4 inches in length (approx.).
c. Moat shall be filled with a braid to insure continuous braid contact and yet permit firm seating of the cavity on the rf cathode terminal contact surface.
d. Location of A-N connector - UG-290/U (two connectors 180 degrees apart).
e. Two holes. Drill and tap for No. 3-48 screws.
f. Three holes. Drill and tap for No. 3-48 screws by 3/16-inch deep.
g. Cylindrical holes H_1 to H_6 have axes coincident within .001 inch.
h. Dimensions are in inches.

FIGURE 4. Input cavity required for resonant frequency test.

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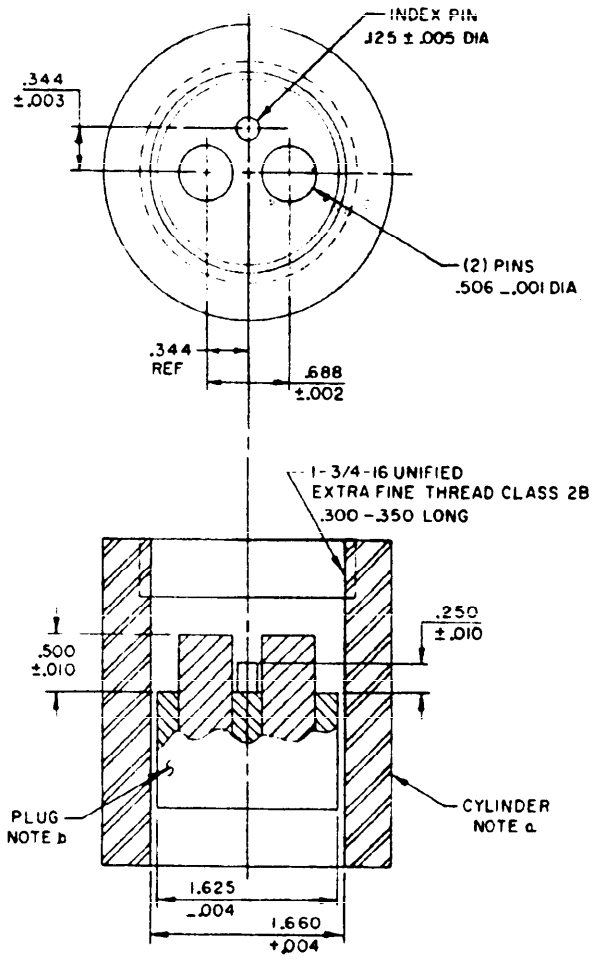
TABLE II. Sampling for examination of packaging



NOTES:
a. Axis of the cylindrical holes and threaded section shall be coincident within .002 inch.
b. Plug sides and pin sides shall be parallel within .001 inch.
c. Dimensions are in inches. Tolerances shown are gage tolerances and shall not be considered tube tolerances.

FIGURE 5. Gage G-1.

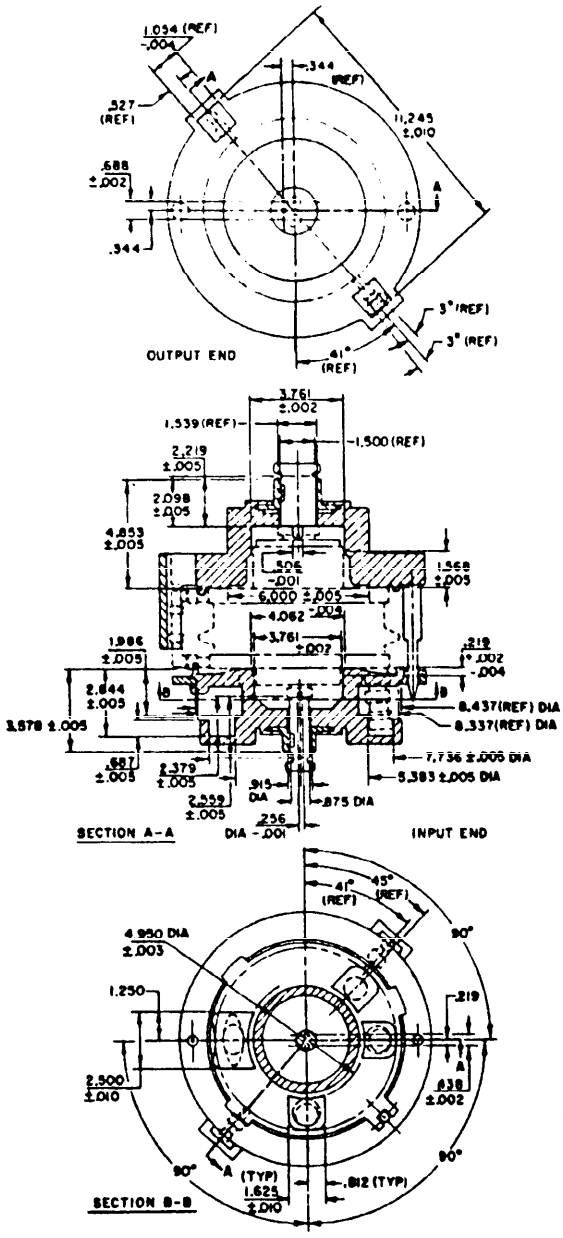
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NOTES:
a. Axis of the cylinder holes and threaded section shall be coincident within .002 inch.
b. Plug sides and pin sides shall be parallel within .001 inch.
c. Dimensions are in inches. Tolerances shown are gage tolerances and shall not be considered tube tolerances.

FIGURE 6. Gage-G-2.

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NOTES:
a. Dimensions are in inches.
b. Tolerances shown are gage tolerances and shall not be considered tube tolerances.
c. See DESC Drawing No. 67038.

FIGURE 7. Gage G-3.

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